

Introduction to Data Science

IN5148: Statistics and Data Science with Applications in Engineering

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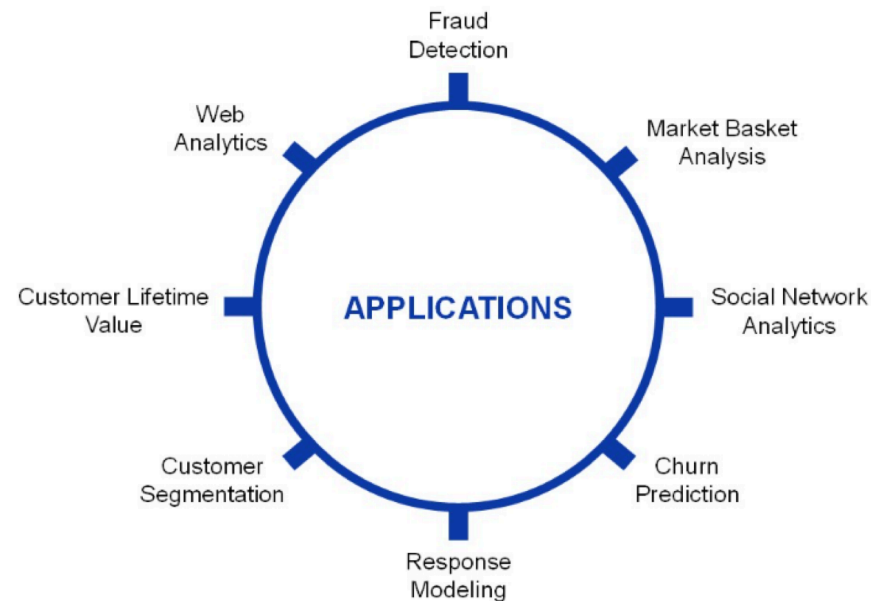
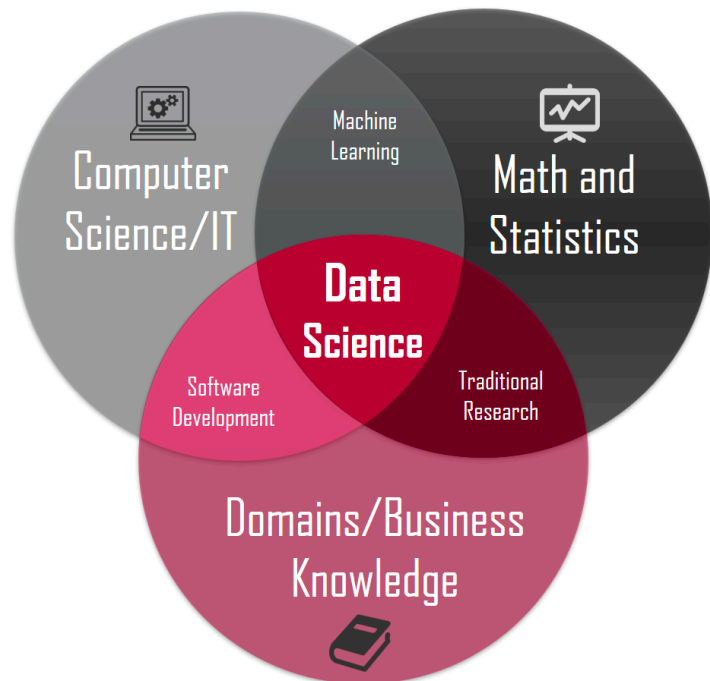
Agenda

1. Introduction to data science
2. Basic notation and terminology

Introduction to data science

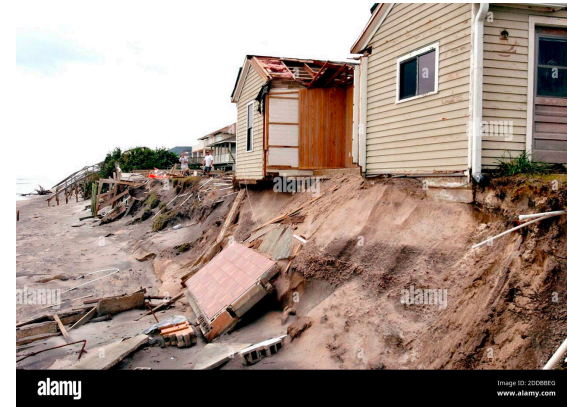
Data science is ...

a multidisciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from vast amounts of structured and unstructured data.



In 2004 ...

Hurricane Frances was sweeping through the Caribbean and threatening to make a direct hit on Florida's Atlantic coast.

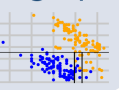


Residents headed for higher ground, but in Arkansas, Wal-Mart executives saw a big opportunity for one of their newest data-driven weapons: ***predictive technology.***

A week before the storm made landfall, Linda M. Dillman, Wal-Mart's chief information officer, pressed her staff to create forecasts based on what had happened when Hurricane Charley hit several weeks earlier.



Backed by trillions of bytes of shopper history stored in Wal-Mart's data warehouse, she said, the company could “start predicting what’s going to happen, rather than waiting for it to happen,” as she put it.



The result

The New York Times reported

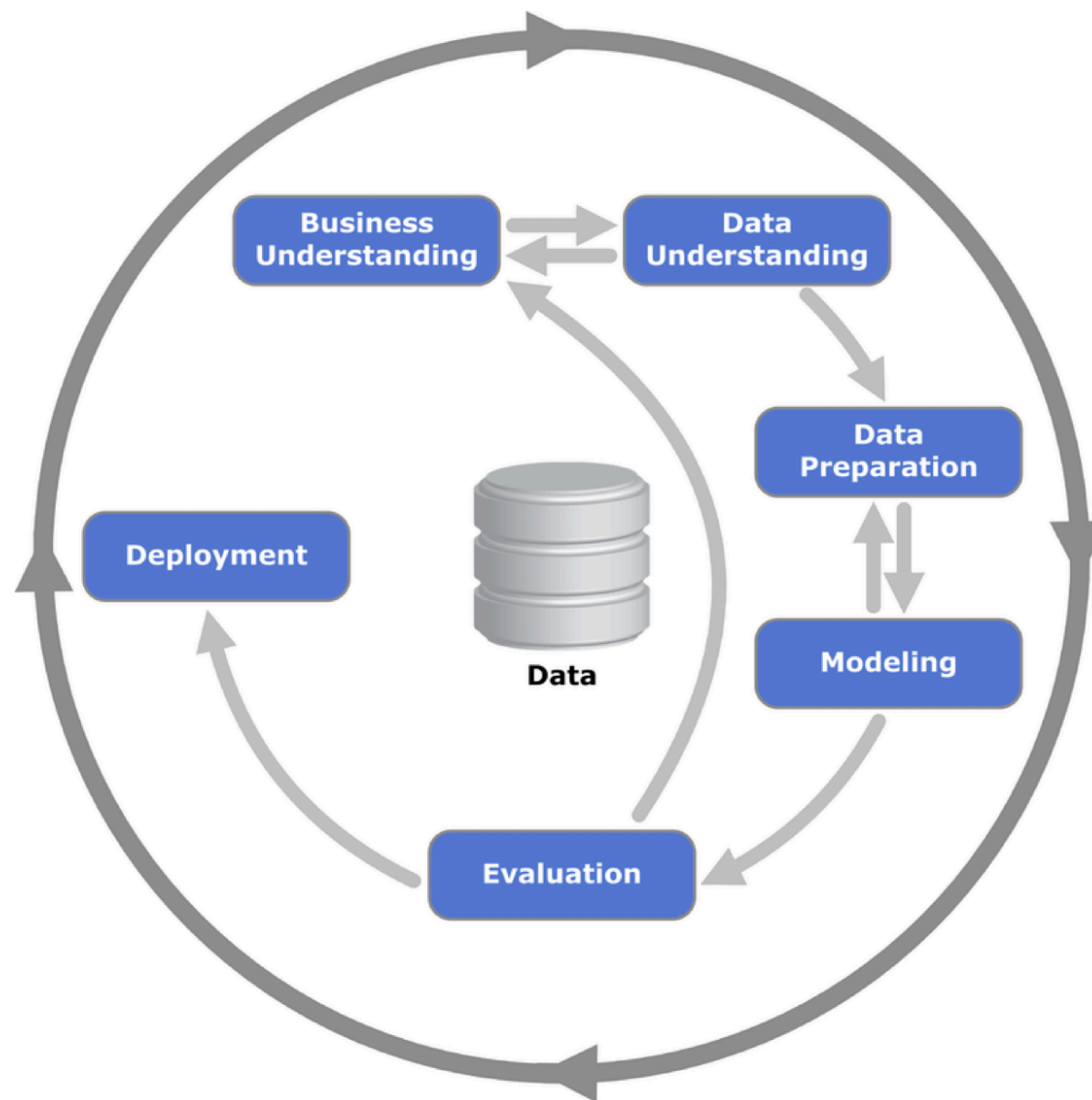
“... Experts analyzed the data and found that stores would indeed need certain products, and not just the usual flashlights.”

Dillman said

“We didn’t know in the past that strawberry Pop-Tarts increase their sales, like seven times their normal sales rate, before a hurricane.”



The scheme of data science



Business understanding

- Business understanding refers to defining the business problem to be solved.
- The goal is to reframe the business problem as a data science problem.
- Often, reframing the problem and designing a solution is an iterative process.

Common data science problems

Regression attempts to estimate or predict, for each individual, the numerical value of some variable for that individual. For example, “How much will a given customer use the service?”

Classification (or class probability estimation) attempts to predict, for each individual in a population, which of a (small) set of classes this individual belongs to. For example, “Among all customers of T-Mobile, which are likely to respond to a given offer?”

Other problems

Clustering attempts to group individuals in a population together by their similarity, but not driven by any specific purpose. For example, “Do our customers form natural groups or segments?”

Discussion

- Often, recasting the problem and designing a solution is an iterative process.
- The initial formulation may not be complete or optimal, so multiple iterations may be necessary for an acceptable solution formulation.
- The key to a great success is a creative problem formulation by some analyst regarding how to cast the business problem as one or more data science problems.

Data understanding I

- If the goal is to solve a business problem, the data that makes up the raw material available from which the solution will be built.
- The available data rarely matches the problem.
- For example, historical data is often collected for purposes unrelated to the current business problem or for no explicit purpose at all.

Data understanding II

- The costs of data vary. Some data will be available for free while others will require effort to obtain.
- A critical part of the data understanding phase is estimating the costs and benefits of each data source and deciding whether further investment is merited.
- Even after all datasets are acquired, collating them may require additional effort.

Our goal

Our goal is to turn data into information that answers useful questions.

Data classes

Text



National Park Service

@NatlParkService

Our favorite part of spring is walking into spider webs and screaming every time. What's yours?

1:10 PM · Mar 24, 2023 · 2M Views

3,355 Retweets

465 Quotes

34.2K Likes

205 Bookmarks



Tweet your reply

Reply



National Park Service

@NatlParkService · Mar 24

Replying to @NatlParkService

We've all walked through a web. Here's a few tips to get through it.

Well, you already went through it, but we digress.

6

63

2,078

68.9K



National Park Service

@NatlParkService · Mar 24

As you walk down a trail, use a stick to make first contact with potential webs, unless you prefer using your face. Do what feels right.

31

162

3,384

147.7K



National Park Service

@NatlParkService · Mar 24

Sleep in and let the early risers clear the trail for you.

Or bring a (We're not going to say it. Rhymes with wall) friend with you and put them up front.

30

91

3,013

139.8K



National Park Service

@NatlParkService · Mar 24

Keep in mind the average spider can spin or repair a web in about 30 to 60 minutes. Something to consider if the trail is a loop or if you're feeling guilty for ruining the spider's day. How could you?

19

110

2,659

158.2K



National Park Service

@NatlParkService · Mar 24

National parks offer many opportunities to hit the trail.

Find the trail for you at:



nps.gov

Trails & Hiking (U.S. National Park Service)

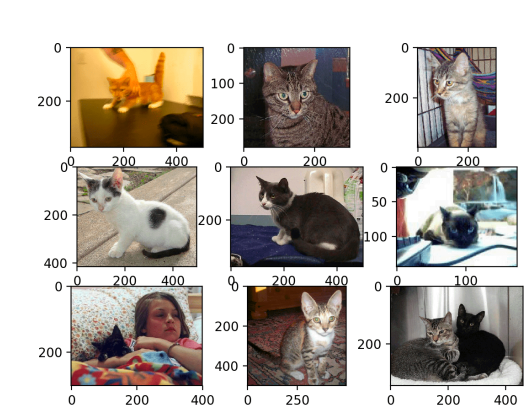
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37

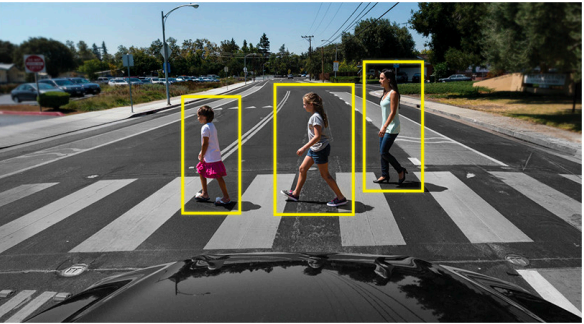
956

100.5K

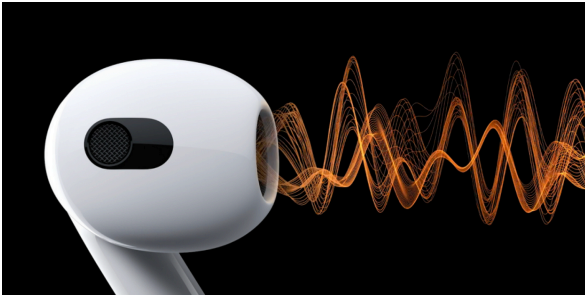
Images



Video

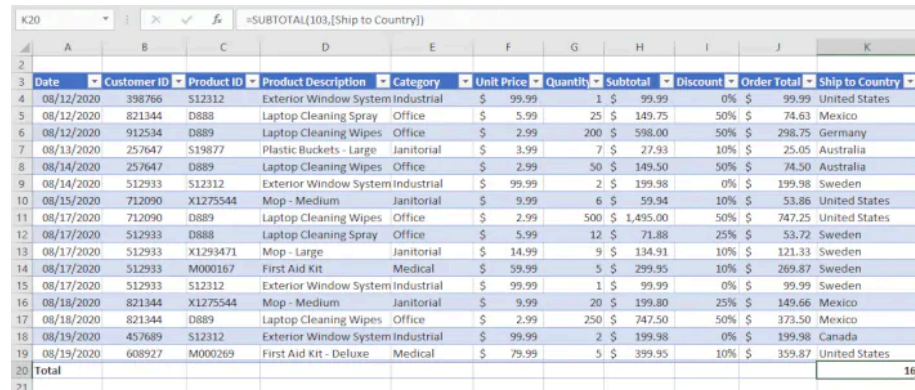


Audio



Numerical data

Data science methodology is based on numerical data given in tables.



	A	B	C	D	E	F	G	H	I	J	K
	Date	Customer ID	Product ID	Product Description	Category	Unit Price	Quantity	Subtotal	Discount	Order Total	Ship to Country
4	08/12/2020	398766	S12312	Exterior Window System	Industrial	\$ 99.99	1	\$ 99.99	0%	\$ 99.99	United States
5	08/12/2020	821344	D888	Laptop Cleaning Spray	Office	\$ 5.99	25	\$ 149.75	50%	\$ 74.63	Mexico
6	08/12/2020	912534	D889	Laptop Cleaning Wipes	Office	\$ 2.99	200	\$ 598.00	50%	\$ 298.75	Germany
7	08/13/2020	257647	S19877	Plastic Buckets - Large	Janitorial	\$ 3.99	7	\$ 27.93	10%	\$ 25.05	Australia
8	08/14/2020	257647	D889	Laptop Cleaning Wipes	Office	\$ 2.99	50	\$ 149.50	50%	\$ 74.50	Australia
9	08/14/2020	512933	S12312	Exterior Window System	Industrial	\$ 99.99	2	\$ 199.98	0%	\$ 199.98	Sweden
10	08/15/2020	712090	X1275544	Mop - Medium	Janitorial	\$ 9.99	6	\$ 59.94	10%	\$ 53.86	United States
11	08/17/2020	712090	D889	Laptop Cleaning Wipes	Office	\$ 2.99	500	\$ 1,495.00	50%	\$ 747.25	United States
12	08/17/2020	512933	D888	Laptop Cleaning Spray	Office	\$ 5.99	12	\$ 71.88	25%	\$ 53.72	Sweden
13	08/17/2020	512933	X1293471	Mop - Large	Janitorial	\$ 14.99	9	\$ 134.91	10%	\$ 121.33	Sweden
14	08/17/2020	512933	M000167	First Aid Kit	Medical	\$ 59.99	5	\$ 299.95	10%	\$ 269.87	Sweden
15	08/17/2020	512933	S12312	Exterior Window System	Industrial	\$ 99.99	1	\$ 99.99	0%	\$ 99.99	Sweden
16	08/18/2020	821344	X1275544	Mop - Medium	Janitorial	\$ 9.99	20	\$ 199.80	25%	\$ 149.66	Mexico
17	08/18/2020	821344	D889	Laptop Cleaning Wipes	Office	\$ 2.99	250	\$ 747.50	50%	\$ 373.50	Mexico
18	08/19/2020	457689	S12312	Exterior Window System	Industrial	\$ 99.99	2	\$ 199.98	0%	\$ 199.98	Canada
19	08/19/2020	608927	M000269	First Aid Kit - Deluxe	Medical	\$ 79.99	5	\$ 399.95	10%	\$ 359.87	United States
20	Total										

In fact, texts, images, videos or audios are transformed into this format to process them.

In this course, we will assume that the data is in a table.

Basic notation and terminology

Notation and terminology

Explanatory variables or predictors:

- X represents an explanatory variable or predictor.
- $\mathbf{X} = (X_1, X_2, \dots, X_p)$ represents a whole collection of p predictors.

Outcome or response:

- Y represents the response variable, which we'll try to predict.

Two main problems

Regression problems. The response Y is quantitative. For example, a person's income, the value of a house, the blood pressure of a patient.

Classification problems. The response Y is qualitative and has K different categories. For example, the brand of a product purchased (A, B, C), or whether a person defaults on a debt (yes or no).

The predictors ($\mathbf{X}$) can be qualitative or quantitative.

Example

What variables predict the presence of Type II diabetes on a person?

- Y is a 1 if a person has Type II diabetes, a 0 if not.
- The predictors ($\mathbf{X}$) might include: income, zip code, age, weight, height, gender and race.

A statistical model

A **statistical model** is a mathematical equation that embodies statistical assumptions concerning the generation of *data*.

Technically, it has the following form:

$$Y = f(\mathbf{X}) + \epsilon$$

where Y is a quantitative response, $f(\mathbf{X})$ is the function that relates the predictors, $\mathbf{X}$, to the Y , and ϵ is the (random) error term.

True and estimated models

$f(\mathbf{X})$ represents the TRUTH. The true relationship between $\mathbf{X}$ and Y .

- Unknown
- Very complex

$\hat{f}(\mathbf{X})$ represents an approximation or estimate of the true model constructed using data.

- Ideally, interpretable (but not necessarily)

Two datasets

- “Training” data are data used to construct $\hat{f}(\mathbf{X})$.
- “Testing” data are data that were NOT used in the fitting process, but are used to test how well your model performs on unseen data.

Yogi Berra

It's tough to make predictions, especially about the future.

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